gpeyt1-21 · Borrow quantity

GEO PE · Long Calculation Problems I — Topic 5: Earth Structures, Ground Improvement & Pavements

Problem

A highway embankment needs 85,000 yd³ of compacted fill at 95% of standard Proctor γd,max = 118.5 pcf. The borrow pit soil has an in-place moist unit weight of 112 pcf at w = 14%. The loose (truck) unit weight is 96 pcf and each truck carries 14 loose yd³. Ignore haul losses.

Most nearly, the volume of borrow (bank measure) required is:

A. 85,436 yd³
B. 102,524 yd³
C. 74,181 yd³
D. 97,397 yd³

Given equations / data

> Dry solids are conserved: Vborrow = Vfill × γd,fill / γd,borrow; γd = γ/(1 + w)

Worked solution

1. γd,fill = 0.95 × 118.5 = 112.6 pcf
2. γd,borrow = 112/1.14 = 98.2 pcf
3. Vborrow = 85,000 × 112.6/98.2 = 97,397 yd³

Final answer

97,397 yd³

Why the other values are wrong

- **85,436 yd³** — used moist borrow unit weight as dry
- **102,524 yd³** — forgot the 95% factor
- **74,181 yd³** — inverted the density ratio

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.8.3 Weight-Volume Relationships, p. 133; §2.3.1 earthwork unit weights (γB, γC)

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gpeyt1-22 · Infinite slope with pore pressure

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Problem

A long natural hillside of residual silty sand slopes at β = 18°. A shallow slip surface parallel to the slope is suspected at z = 8 ft (vertical depth). γ = 122 pcf, c′ = 0, φ′ = 30°. During wet season a piezometer on the slip surface reads u = 280 psf. The road at the toe carries 2,000 vehicles/day.

Most nearly, the factor of safety during the wet season is:

A. 1.78
B. 1.27
C. 1.21
D. 0.68

Given equations / data

> $r_u = u/(\gamma z)$; $F = [1 - r_u(1 + \tan^2\beta)] \tan\phi'/\tan\beta$

Worked solution

1. $r_u = 280/(122 \times 8) = 0.287$
2. $1 + \tan^2 18 = 1.1056$
3. $F = [1 - 0.287 \times 1.1056] \times \tan 30/\tan 18 = 0.6828 \times 1.777 = 1.21$

Final answer

1.21

Why the other values are wrong

- **1.78** — dry case (ignored u)
- **1.27** — omitted the (1 + tan²β) term
- **0.68** — used γw instead of γ in ru

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.6.4 Infinite Slope, p. 105

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gpeyt1-23 · Flexible pavement SN

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Problem

An AASHTO 1993 flexible pavement design gives a required structural number SN = 4.6 for 8.2 million ESALs (R = 95%, S0 = 0.45, ΔPSI = 1.7, subgrade MR = 6,000 psi). Layers: 4.0 in. HMA (a1 = 0.44); crushed-stone base (a2 = 0.14, m2 = 1.0); 8.0 in. granular subbase (a3 = 0.11, m3 = 0.9).

Most nearly, the minimum base thickness D2 is:

A. 14.0 in.
B. 14.6 in.
C. 20.3 in.
D. 16.3 in.

Given equations / data

```
SN = a1D1 + a2D2m2 + a3D3m3
```

Worked solution

1. a1D1 = 0.44×4.0 = 1.76
2. a3D3m3 = 0.11×8.0×0.9 = 0.792
3. D2 = (4.6 − 1.76 − 0.792)/(0.14×1.0) = 14.6 in. (round up in practice)

Final answer

14.6 in.

Why the other values are wrong

- **14.0 in.** — ignored the subbase drainage coefficient m3
- **20.3 in.** — forgot the subbase contribution
- **16.3 in.** — applied m3 = 0.9 to the base instead

References (Jayce's library)

- FHWA NHI-05-037 Geotechnical Aspects of Pavements, Eq. 5.40 (SN = a1D1 + a2D2m2 + a3D3m3); NCEES PE Civil Reference Handbook v2.1 §3.19 Pavements

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=uGGy0dkZaX4, https://www.youtube.com/watch?v=wBgMPZkg_IE (no creator values copied).

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gpeyt1-24 · Geotextile wall spacing

GEO PE · Long Calculation Problems I — Topic 5: Earth Structures, Ground Improvement & Pavements

Problem

A 16 ft geotextile-wrapped reinforced wall retains granular backfill (γ = 120 pcf, φ = 32°). A uniform surcharge q = 250 psf acts on top (no live point loads). The geotextile has wide-width Tult = 2,400 lb/ft and cumulative reduction factors ΠRF = 2.6. Use FS = 1.5 on reinforcement. Check the bottom layer at Z = 16 ft. Facing batter is vertical; foundation is dense gravel.

Most nearly, the maximum vertical spacing of the bottom layer is:

A. 2.40 ft
B. 1.04 ft
C. 0.09 ft
D. 0.92 ft

Given equations / data

```
σh = Ka γ Z + Ka q; Tallow = Tult/ΠRF; Sv = Tallow/(σh · FS); Ka = tan²(45 − φ/2)
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Worked solution

1. Ka = tan²(45 − 32/2) = 0.307
2. σh = 0.307×120×16 + 0.307×250 = 667 psf
3. Tallow = 2,400/2.6 = 923 lb/ft
4. Sv = 923/(667×1.5) = 0.92 ft (≈ 11 in.)

Final answer

0.92 ft

Why the other values are wrong

- **2.40 ft** — used Tult without reduction factors
- **1.04 ft** — omitted the surcharge pressure
- **0.09 ft** — used Kp in place of Ka

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.13.5.1 Geotextile-Reinforced Walls, p. 170

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gpeyt1-25 · Geotextile length

GEO PE · Long Calculation Problems I — Topic 5: Earth Structures, Ground Improvement & Pavements

Problem

Same 16 ft geotextile wall: $\gamma = 120$ pcf, $\varphi = 32°$, $q = 250$ psf, FS = 1.5. Design a layer at $Z = 4$ ft below the top with $S_v = 2.0$ ft. Interface friction $\delta = \tfrac{2}{3}\varphi$, $c_a = 0$. Wall length $L = L_e + L_R$ (L_R measured from the face to the Rankine plane).

Most nearly, the total geotextile length L at Z = 4 ft is:

A. 1.8 ft
B. 10.2 ft
C. 4.0 ft
D. 8.4 ft

Given equations / data

$$\sigma_h = K_a(\gamma Z + q);\quad L_e = \frac{S_v\,\sigma_h\,FS}{2(c_a + \gamma Z \tan\delta)};\quad L_R = (H - Z)\tan\left(45 - \frac{\varphi}{2}\right)$$

Worked solution

1. $\sigma_h = 0.307(120\times4 + 250) = 224$ psf
2. $\delta = 21.33°$; $L_e = \dfrac{2.0\times224\times1.5}{2\times120\times4\times\tan21.33} = 1.79$ ft
3. $L_R = (16 - 4)\tan(45 - 16) = 6.65$ ft
4. $L = 1.79 + 6.65 = 8.4$ ft (a practical minimum length would also be checked)

Final answer

8.4 ft

Why the other values are wrong

- **1.8 ft** — reported L_e only (forgot L_R)
- **10.2 ft** — used one face of friction (dropped the 2)
- **4.0 ft** — measured L_R with Z instead of $H - Z$

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.13.5.1 Geotextile-Reinforced Walls, p. 170

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